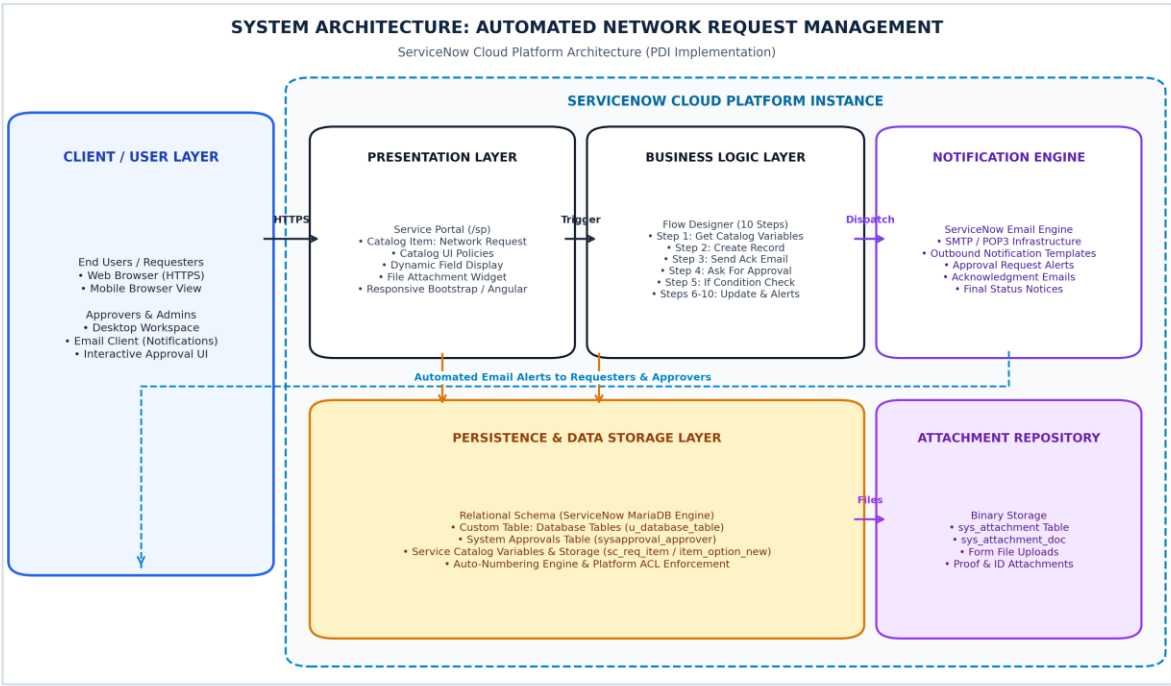


Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	03 October 2026
Developer / Student	Vinay Krishna A K
Project Name	Automated Network Request Management in ServiceNow

1. Technical Architecture



Architecture Overview: The solution is implemented on the ServiceNow Cloud Platform utilizing a multi-tier architectural pattern. The client presentation tier leverages the Service Portal (/sp) with dynamic Catalog UI Policies for request intake. The application logic tier executes a 10-step Flow Designer workflow to manage approvals, database persistence, and branching. Data persistence is handled by a custom relational table Database Tables (u_database_table) backed by ServiceNow's cloud database engine, while platform notification services handle automated outbound communications.

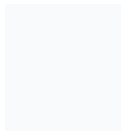
Table 1: Technology Stack Components

S.No	Component	Description	Technology / Implementation
1.	User Interface	Self-service intake form on the Service Portal (/sp) with dynamic fields, choice options, and attachment widget.	ServiceNow Service Portal, AngularJS, HTML5, CSS3, Bootstrap
2.	Application Logic 1 (Intake & UI Policies)	Dynamic client-side form controls to show/hide fields, validate mandatory fields, and handle connection types.	ServiceNow Catalog UI Policies & Client Scripts (JavaScript)
3.	Application Logic 2 (Workflow Engine)	10-step low-code automation workflow handling variable extraction, record creation, approval routing, and branching.	ServiceNow Flow Designer (Trigger: Service Catalog)
4.	Application Logic 3 (Approval Routing)	Automated approval gate routing requests to designated Manager/Admin and updating approval states.	Flow Action: Ask For Approval (sysapproval_approver)
5.	Database (Custom Table)	Dedicated custom relational table storing all request variables, payment options, addresses, and states.	Database Tables (u_database_table) with Auto-Numbering
6.	Cloud Database	Enterprise relational database backend managed natively by the ServiceNow cloud infrastructure.	ServiceNow Managed Relational Cloud DB (MariaDB backend)
7.	File Storage	Secure repository storing documents and proof files uploaded via the portal form.	ServiceNow Attachment Architecture (sys_attachment / sys_attachment_doc)
8.	Notification Engine	Automated outbound notification services generating emails for acknowledgment, approval alerts, and closure.	ServiceNow Email Notification Engine (SMTP / Flow Send Email)

9.	Security & Access Controls	Role-based security controlling read, write, create, and delete permissions on records and forms.	Access Control Lists (ACLs) & User Roles (snc_internal, admin)
10.	Infrastructure (Platform / Cloud)	Multi-tenant enterprise cloud environment hosting the personal developer instance (PDI).	ServiceNow Enterprise Cloud Infrastructure (SaaS / PaaS)

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology / Mechanism
1.	Open-Source & Standard Frameworks	Underlying front-end responsive web frameworks powering the Service Portal and UI components.	AngularJS, Bootstrap 3.x, HTML5, CSS3, JavaScript (ES6)
2.	Security Implementations	Multi-layer defense: HTTPS (TLS 1.3) data encryption in transit, AES-256 at rest, role-based ACLs, and strict input sanitization against XSS/injection.	Platform ACLs, TLS 1.3, AES-256 Encryption, Role-Based Access Control (RBAC)
3.	Scalable Architecture	Multi-instance cloud architecture decoupling client portal rendering, workflow processing, and database persistence to support growing transaction volumes.	ServiceNow Advanced Multi-Instance Architecture & Load Balancers
4.	Availability	Enterprise-grade 24/7 uptime with automated data replication across paired geographic data centers and high-availability clustering.	ServiceNow High-Availability Data Center Pairs & Automatic Failover
5.	Performance	Sub-second response times achieved via native portal caching, asynchronous Flow Designer	In-memory Cache (GlideCache), Asynchronous Flows,



execution, and indexing on custom DB B-Tree Indexes
table search fields.